The Scope and Method of Economics

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The study of economics should begin with a sense of wonder. Pause for a moment and consider a typical day in your life. It might start with a bagel made in a local bakery with flour produced in Minnesota from wheat grown in Kansas. After class you drive with a friend on an interstate highway that is part of a system that took 20 years and billions of dollars to build. You stop for gasoline refined in Louisiana from Saudi Arabian crude oil. Later, you log onto the Web with a laptop assembled in Indonesia from parts made in China and Skype with your brother in Mexico City. You use or consume tens of thousands of things in a day. Someone organized men and women and materials to produce and distribute these things. Thousands of decisions went into their completion, and somehow they got to you.

In the United States, more than 160 million people—over half the total population—work at hundreds of thousands of different jobs producing more than $18 trillion worth of goods and services every year. Some cannot find work; some choose not to work. The United States imports more than $300 billion worth of automobiles and parts and more than $350 billion worth of petroleum and petroleum products each year; it exports around $140 billion worth of agricultural products, including food. In the modern economy, consumers' choices include products made all over the globe.

Economics is the study of how individuals and societies choose to use the scarce resources that nature and previous generations have provided. The key word in this definition is *choose*. Economics is a behavioral, or social, science. In large measure, it is the study of how people make choices. The choices that people make, when added up, translate into societal choices.

The purpose of this chapter and the next is to elaborate on this definition and to introduce the subject matter of economics. What is produced? How is it produced? Who gets it? Why? Is the result good or bad? Can it be improved?

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economics The study of how individuals and societies choose to use the scarce resources that nature and previous generations have provided.

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1.1 LEARNING OBJECTIVE

Identify three key reasons to study economics. Think of an example from your life in which understanding opportunity costs or the principle of efficient markets could make a difference in your decision making.

Why Study Economics?

There are three main reasons to study economics: to learn a way of thinking, to understand society, and to be an informed citizen.

To Learn a Way of Thinking MyLab Economics Concept Check

Probably the most important reason for studying economics is to learn a way of thinking. Economics has three fundamental concepts that, once absorbed, can change the way you look at everyday choices: opportunity cost, marginalism, and the working of efficient markets.

Opportunity Cost

What happens in an economy is the outcome of thousands of individual decisions. People must decide how to divide their incomes among all the goods and services available in the marketplace. They must decide whether to work, whether to go to school, and how much to save. Businesses must decide what to produce, how much to produce, how much to charge, and where to locate. Economic analysis provides a structured way of thinking about these types of decisions.

Nearly all decisions involve trade-offs. A key concept that recurs in analyzing the decision-making process is the notion of *opportunity cost*. The full "cost" of making a specific choice includes what we give up by not making the best alternative choice. The best alternative that we forgo, or give up, when we make a choice or a decision is called the **opportunity cost** of that decision.

opportunity cost The best alternative that we forgo, or give up, when we make a choice or a decision.

When asked how much a movie costs, most people cite the ticket price. For an economist, this is only part of the answer: to see a movie takes not only a ticket but also time. The opportunity cost of going to a movie is the value of the other things you could have done with the same money and time. If you decide to take time off from work, the opportunity cost of your leisure is the pay that you would have earned had you worked. Part of the cost of a college education is the income you could have earned by working full time instead of going to school.

scarce Limited.

Opportunity costs arise because resources are scarce. **Scarce** simply means limited. Consider one of our most important resources—time. There are only 24 hours in a day, and we must live our lives under this constraint. A farmer in rural Brazil must decide whether it is better to continue to farm or to go to the city and look for a job. A hockey player at the University of Vermont must decide whether to play on the varsity team or spend more time studying. In the *Economics in Practice* box on page 29, we use the idea of opportunity cost to help explain how rainfall in India affects math scores of rural children. As you will see, opportunity cost is a powerful idea.

Marginalism

A second key concept used in analyzing choices is the notion of **marginalism**. In weighing the costs and benefits of a decision, it is important to weigh only the costs and benefits that arise from the decision. Suppose, for example, that you live in New Orleans and that you are weighing the costs and benefits of visiting your mother in Iowa. If business required that you travel to Kansas City anyway, the cost of visiting Mom would be only the additional, or *marginal*, time and money cost of getting to Iowa from Kansas City.

marginalism The process of analyzing the additional or incremental costs or benefits arising from a choice or decision.

There are numerous examples in which the concept of marginal cost is useful. For an airplane that is about to take off with empty seats, the marginal cost of an extra passenger is essentially zero; the total cost of the trip is roughly unchanged by the addition of an extra passenger. Thus, setting aside a few seats to be sold at big discounts through www.priceline.com or other Web sites can be profitable even if the fare for those seats is far below the average cost per seat of making the trip. As long as the airline succeeds in filling seats that would otherwise have been empty, doing so is profitable.

Efficient Markets—No Free Lunch

Suppose you are ready to check out at a busy grocery store on the day before a storm and seven checkout registers are open with several people in each line. Which line should you choose? Clearly you should go to the shortest line! But if everyone thinks this way—as is likely—all the lines will be equally long as people move around. Economists often loosely refer to "good deals" or risk-free ventures as *profit opportunities*. Using the term loosely, a profit opportunity exists at the checkout lines when one line is shorter than the others. In general, such profit opportunities are rare. At any time, many people are searching for them; as a consequence, few exist. Markets like this, where any profit opportunities are eliminated almost

ECONOMICS IN PRACTICE

Rainfall and Schooling in India

As we indicated in the text, the idea of opportunity cost is one of the fundamental concepts in economics. When we look at the choices people make in the area of employment and education, the role of opportunity cost is especially large. Recent work looking at the effect of rainfall on children's education in India highlights the role that opportunity cost can play.[1]

Much of India is still rural and dependent on agriculture. Most adults, both male and female, are engaged in agriculture, and in most families the children also play a role in agricultural production. Irrigation is uncommon, especially in the poorer areas of India, and as a result agricultural production is highly dependent on rainfall. When rains are unusually plentiful, not only are harvests larger, but the gains from having people work the land increase. In a drought there is very little a farm worker can do to increase yields, and there is little produce to harvest. It follows then that when rains are unusually plentiful in an area, the opportunity cost of having someone out of the labor force increases.

Think for a moment about families with children, choosing between sending them to school, which would make them more productive in their later life, or sending them to the fields to help with the current harvest. The opportunity cost of sending your children to school is the loss in current agricultural output. If there have been ample rains, that opportunity cost is high. In a drought, the cost is low.

It follows from this opportunity cost differential that one would expect fewer children at school when the rains have been plentiful in rural India than in a drought. This is precisely what Shah and Steinberg find. Using data from more than 2 million children ages 5–16 across rural India, these economists find that an unusually high rainfall reduces school



enrollments by a significant amount. And, unsurprisingly, these children end up with significantly lower math scores on tests administered by the state. You should be able to see the power of the concept of opportunity cost. In this example, it allows us to see the effect of rainfall on rural math scores.

CRITICAL THINKING

1. For urban children in India, work opportunities are few. What would you expect to see happen to the urban-rural gap in test scores in high rainfall periods?

[1] Manisha Shah and Bryce Millett Steinberg, "Drought of Opportunities: Contemporaneous and Long Term Impacts of Rainfall Shocks on Human Capital" *Journal of Political Economy*, April 2017, 527–561.

instantaneously, are said to be **efficient markets**. (We discuss *markets*, the institutions through which buyers and sellers interact and engage in exchange, in detail in Chapter 2.)

efficient market A market in which profit opportunities are eliminated almost instantaneously.

The common way of expressing the efficient markets concept is "there's no such thing as a free lunch." How should you react when a stockbroker calls with a hot tip on the stock market? With skepticism. Thousands of individuals each day are looking for hot tips in the market. If a particular tip about a stock is valid, there will be an immediate rush to buy the stock, which will quickly drive up its price. This view that very few profit opportunities exist can, of course, be carried too far. There is a story about two people walking along, one an economist and one not. The non-economist sees a $20 bill on the sidewalk and says, "There's a $20 bill on the sidewalk." The economist replies, "That is not possible. If there were, somebody would already have picked it up."

There are clearly times when profit opportunities exist. Someone has to be first to get the news, and some people have quicker insights than others. Nevertheless, news travels fast, and there are thousands of people with quick insights. The general view that large profit opportunities are rare is close to the mark and is powerful in helping to guide decision making. The *Economics in Practice* box on page 30 describes the way in which learning this way of thinking can pay off in labor market outcomes.

The study of economics teaches us a way of thinking and helps us make decisions.

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ECONOMICS IN PRACTICE

Majoring in Economics Makes You Less Vulnerable to a Recession!

It is well known that a college education, on average, increases one's income. Economists estimate that over one's lifetime, a college degree holder will earn on average almost 70 percent more than someone with only a high school degree. Part of the returns to a college education come from higher wages and part from being less likely to suffer long spells of unemployment. It is perhaps less well known that both wage and unemployment effects also vary considerably with the majors of college graduates. Economics is, along with engineering, one of the majors with the highest wage premia.

Recent work has shown yet another advantage of the economics major: It helps to protect graduates from the long-term effects of graduating in a recession.[1] As Lisa Kahn found in some of her earlier work, graduating in a recession (a period of high unemployment and low economic growth) has long-term negative effects on one's career. One's first job under these circumstances tends to be worse than otherwise, and this bad placement affects the next few job opportunities and hence one's lifetime earnings. But Kahn's recent work suggests that the extent of this long-term recession handicap varies considerably with one's major. Majors like economics are less hurt by graduating in a recession than sociology or journalism, for example. Learning to think like an economist not only generates a higher wage but provides insurance against volatility in the economy!



CRITICAL THINKING

1. Why does a recent graduate's first job matter for his or her long-term earnings, even if he or she only stays at that job for three years?

[1] Joseph Altonji, Lisa Kahn, Jamin Speer, "Cashier or Consultant? Entry Labor Market Conditions, Field of Study and Career Success." *Journal of Labor Economics*, 2016, (34) S361–S401.

To Understand Society MyLab Economics Concept Check

Another reason for studying economics is to understand society better. Past and present economic decisions have an enormous influence on the character of life in a society. The current state of the physical environment, the level of material well-being, and the nature and number of jobs are all products of the economic system.

Industrial Revolution The period in England during the late eighteenth and early nineteenth centuries in which new manufacturing technologies and improved transportation gave rise to the modern factory system and a massive movement of the population from the countryside to the cities.

At no time has the impact of economic change on a society been more evident than in England during the late eighteenth and early nineteenth centuries, a period that we now call the **Industrial Revolution**. Increases in the productivity of agriculture, new manufacturing technologies, and development of more efficient forms of transportation led to a massive movement of the British population from the countryside to the city. At the beginning of the eighteenth century, approximately 2 out of 3 people in Great Britain worked in agriculture. By 1812, only 1 in 3 remained in agriculture; by 1900, the figure was fewer than 1 in 10. People jammed into overcrowded cities and worked long hours in factories. England had changed completely in two centuries—a period that in the run of history was nothing more than a blink of an eye.

The discipline of economics began to take shape during this period. Social critics and philosophers looked around and knew that their philosophies must expand to accommodate the changes. Adam Smith's *Wealth of Nations* appeared in 1776. It was followed by the writings of David Ricardo, Karl Marx, Thomas Malthus, and others. Each tried to make sense out of what was happening. Who was building the factories? Why? What determined the level of wages paid to workers or the price of food? What would happen in the future, and what *should* happen? The people who asked these questions were the first economists.

Societal changes are often driven by economics. Consider the developments in the early years of the World Wide Web. Changes in the ways people communicate with one another and

with the rest of the world, largely created by private enterprise seeking profits, have affected almost every aspect of our lives, from the way we interact with friends and family to the jobs that we have and the way cities and governments are organized.

The study of economics is an essential part of the study of society.

To Be an Informed Citizen MyLab Economics Concept Check

A knowledge of economics is essential to being an informed citizen. Between 2008 and 2013, much of the world struggled with a major recession and slow recovery, leaving millions of people around the world out of work. Understanding what happens in a recession and what the government can and cannot do to help in a recovery is an essential part of being an informed citizen. In the early years of President Trump's administration, the country grappled with questions of immigration, trade policy, and tax structure. An understanding of economics is fundamental to making national policy in all of these areas.

Economics is also essential in understanding a range of other everyday government decisions at the local and federal levels. Why do governments pay for public schools and roads, but not cell phones? The federal government under President Barack Obama moved toward universal health care for U.S. citizens, while President Trump moved to limit the Affordable Care Act. What are the pros and cons of these policies? In some states, scalping tickets to a ball game is illegal. Is this a good policy or not? Every day, across the globe, people engage in political decision making around questions like these, questions that depend on an understanding of economics.

To be an informed citizen requires a basic understanding of economics.

The Scope of Economics

1.2 LEARNING OBJECTIVE

Describe microeconomics, macroeconomics, and the diverse fields of economics.

Most students taking economics for the first time are surprised by the breadth of what they study. Some think that economics will teach them about the stock market or what to do with their money. Others think that economics deals exclusively with problems such as inflation and unemployment. In fact, it deals with all those subjects, but they are pieces of a much larger puzzle. Economists use their tools to study a wide range of topics.

The easiest way to get a feel for the breadth and depth of what you will be studying is to explore briefly the way economics is organized. First, there are two major divisions of economics: microeconomics and macroeconomics.

Microeconomics and Macroeconomics MyLab Economics Concept Check

Microeconomics deals with the functioning of individual industries and the behavior of individual economic decision-making units: firms and households. Firms' choices about what to produce and how much to charge and households' choices about what and how much to buy help to explain why the economy produces the goods and services it does.

microeconomics The branch of economics that examines the functioning of individual industries and the behavior of individual decision-making units—that is, firms and households.

Another big question addressed by microeconomics is who gets the goods and services that are produced. Understanding the forces that determine the distribution of output is the province of microeconomics. Microeconomics helps us to understand how resources are distributed among households. Recent research has shown an increase in income inequality in the United States. Why has this occurred? What determines who is rich and who is poor?

Macroeconomics looks at the economy as a whole. Instead of trying to understand what determines the output of a single firm or industry or what the consumption patterns are of a single household or group of households, macroeconomics examines the factors that determine national output, or national product. Microeconomics is concerned with *household* income; macroeconomics deals with *national income*.

macroeconomics The branch of economics that examines the economic behavior of aggregates—income, employment, output, and so on—on a national scale.

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Whereas microeconomics focuses on individual product prices and relative prices, macroeconomics looks at the overall price level and how quickly (or slowly) it is rising (or falling). Microeconomics questions how many people will be hired (or fired) this year in a particular industry or in a certain geographic area and focuses on the factors that determine how much labor a firm or an industry will hire. Macroeconomics deals with *aggregate* employment and unemployment: how many jobs exist in the economy as a whole and how many people who are willing to work are not able to find work.

To summarize:

> Microeconomics looks at the individual unit—the household, the firm, the industry. It sees and examines the "trees." Macroeconomics looks at the whole, the aggregate. It sees and analyzes the "forest."

ECONOMICS IN PRACTICE

How Italian Is a Ferrari?

Before we attempt to explain the workings of an economy, it is worthwhile to see the impact of the decisions and operations of individual firms. We need to understand why some companies elect to move part or most of their manufacturing work abroad, even if at the risk of higher shipping costs and manufacturing delays. We need to understand why governments tolerate this behavior even if it deprives their economies of jobs and income.

No nation can operate as a closed economic system that produces all goods or resources necessary for the production of goods and services. Since early times, various nations of the world have exchanged resources and products through trade. But to defend their economies from foreign competition, governments have been protecting domestic industries using tariffs and quotas on imports as well as with restrictions on exporting essential capital equipment and factors of production. The peacekeeping aspiration after World War II made it imperative to move the modern world towards a more cooperative and liberalized trade system. Nations started to sign multilateral free trade agreements (FTAs) in tandem with bilateral, local, and regional FTAs. By 2016–2017 the volume of global trade reached nearly €13 trillion, as reported by the World Trade Organization.[1]

The European Union (EU) is the largest single market in the world. EU legislation allows its manufacturers to claim items assembled in their country as their own products. Let us take the case of auto producer Ferrari, the flagship of Italian automobile industry. Ferrari has earned itself the reputation of the leading brand, especially in Formula One (F1) auto racing championships, as it has achieved the highest number of F1 championships and produced the largest number of winning drivers. Some would assume that every single component of Ferrari sports cars is from Italy since it carries the sign "Made in Italy." However, contrary to popular belief, Ferrari parts are produced outside Italy, assembled at the legendary Maranello factory, labeled "Made in Italy," and then either sold in Italy or exported. While most of the engine is manufactured in Italy, Ferrari uses Japanese NGK spark plugs, German Mahle turbo engine components, and



American Goodyear tyres.[2] Domestic protection activists supporting Italian manufacturers have argued that by outsourcing these parts, Ferrari moves jobs from Italy to Japan, Germany, and USA. They claim that the success of companies such as Ferrari comes at the expense of the Italian workforce, which in turn hurts Italy's economy.

As you can see, the line between "made" and "assembled" is rather blurred. Consumers do not mind if domestic firms outsource production to foreign producers as long as they provide better quality products at lower prices. So, the next time you see a label that reads "Made in Italy," keep in mind that from an economics point of view, you may need to dig a little deeper to see what is really going on.

CRITICAL THINKING

1. What are the factors that could drive manufacturers' decisions to outsource manufacturing to foreign countries?

[1] World Trade Organization, 2017. *Annual Report*-2017, Geneva: WTO.
[2] Brand Finance, 2017. "Ferrari Brand Speeds Ahead of the Field," *Brand Finance*.

TABLE 1.1 Examples of Microeconomic and Macroeconomic Concerns

Division of Economics	Production	Prices	Income	Employment
Microeconomics	*Production/output in individual industries and businesses* How much steel How much office space How many cars	*Prices of individual goods and services* Price of medical care Price of gasoline Food prices Apartment rents	*Distribution of income and wealth* Wages in the auto industry Minimum wage Executive salaries Poverty	*Employment by individual businesses and industries* Jobs in the steel industry Number of employees in a firm Number of accountants
Macroeconomics	*National production/output* Total industrial output Gross domestic product Growth of output	*Aggregate price level* Consumer prices Producer prices Rate of inflation	*National income* Total wages and salaries Total corporate profits	*Employment and unemployment in the economy* Total number of jobs Unemployment rate

Table 1.1 summarizes these divisions of economics and some of the subjects with which they are concerned.

The Diverse Fields of Economics MyLab Economics Concept Check

Individual economists focus their research and study in many different areas. The subfields of economics are listed in Table 1.2 along with a sample research or policy question that an economist in this subfield might study.

TABLE 1.2 The Fields of Economics

Behavioral economics	Do aggregate household savings increase when we automatically enroll people in savings programs and let them opt out as opposed to requiring them to sign up?
Comparative economic systems	How does the resource allocation process differ in market versus command and control systems?
Econometrics	What inferences can we make based on conditional moment inequalities?
Economic development	Does increasing employment opportunities for girls in developing nations increase their educational achievement?
Economic history	How did the growth of railroads and improvement in transportation more generally change the U.S. banking systems in the nineteenth century?
Environmental economics	What effect would a tax on carbon have on emissions? Is a tax better or worse than rules?
Finance	Is high frequency trading socially beneficial?
Health economics	Do co-pays by patients change the choice and use of medicines by insured patients?
The history of economic thought	How did Aristotle think about just prices?
Industrial organization	How do we explain price wars in the airline industry?
International economics	What are the benefits and costs of free trade? Does concern about the environment change our views of free trade?
Labor economics	Will increasing the minimum wage decrease employment opportunities?
Law and economics	Does the current U.S. patent law increase or decrease the rate of innovation?
Public economics	Why is corruption more widespread in some countries than in others?
Urban and regional economics	Do enterprise zones improve employment opportunities in central cities?

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1.3 LEARNING OBJECTIVE

Think about an example of bad causal inference leading to erroneous decision making. Identify the four main goals of economic policy.

The Method of Economics

Economics asks and attempts to answer two kinds of questions: positive and normative. **Positive economics** attempts to understand behavior and the operation of economic systems *without making judgments* about whether the outcomes are good or bad. It strives to describe what exists and how it works. What determines the wage rate for unskilled workers? What would happen if the United States substantially lowered the corporate profit tax, as it did in 2018? The answers to such questions are the subject of positive economics.

positive economics An approach to economics that seeks to understand behavior and the operation of systems without making judgments. It describes what exists and how it works.

In contrast, **normative economics** looks at the outcomes of economic behavior and asks whether they are good or bad and whether they can be made better. Normative economics involves judgments and prescriptions for courses of action. Should the government subsidize or regulate the cost of higher education? Should the United States allow importers to sell foreign-produced goods that compete with U.S.-made products? Should we reduce or eliminate inheritance taxes? Normative economics is often called *policy economics*.

normative economics An approach to economics that analyzes outcomes of economic behavior, evaluates them as good or bad, and may prescribe courses of action. Also called *policy economics*.

Of course, most normative questions involve positive questions. To know whether the government *should* take a particular action, we must know first if it *can* and second what the consequences are likely to be.

Theories and Models MyLab Economics Concept Check

In many disciplines, including physics, chemistry, meteorology, political science, and economics, theorists build formal models of behavior. A **model** is a formal statement of a theory. It is usually a mathematical statement of a presumed relationship between two or more variables.

model A formal statement of a theory, usually a mathematical statement of a presumed relationship between two or more variables.

A **variable** is a measure that can change from time to time or from observation to observation. Income is a variable—it has different values for different people and different values for the same person at different times. The price of a quart of milk is a variable; it has different values at different stores and at different times. There are countless other examples.

variable A measure that can change from time to time or from observation to observation.

Because all models simplify reality by stripping part of it away, they are abstractions. Critics of economics often point to abstraction as a weakness. Most economists, however, see abstraction as a real strength.

The easiest way to see how abstraction can be helpful is to think of a map. A map is a representation of reality that is simplified and abstract. A city or state appears on a piece of paper as a series of lines and colors. The amount of reality that the mapmaker can strip away before the map loses something essential depends on what the map will be used for. If you want to drive from St. Louis to Phoenix, you need to know only the major interstate highways and roads. However, to travel around Phoenix, you may need to see every street and alley.

Like maps, economic models are abstractions that strip away detail to expose only those aspects of behavior that are important to the question being asked. The principle that irrelevant detail should be cut away is called the principle of **Ockham's razor**, named after the 14th-century philosopher William of Ockham.

Ockham's razor The principle that irrelevant detail should be cut away.

Be careful: Although abstraction is a powerful tool for exposing and analyzing specific aspects of behavior, it is possible to oversimplify. Economic models often strip away a good deal of social and political reality to get at underlying concepts. When an economic theory is used to help formulate actual government or institutional policy, political and social reality must often be reintroduced if the policy is to have a chance of working.

The appropriate amount of simplification and abstraction depends on the use to which the model will be put. To return to the map example: You do not want to walk around San Francisco with a map made for drivers—there are too many very steep hills.

All Else Equal It is usually true that whatever you want to explain with a model depends on more than one factor. Suppose, for example, that you want to explain the total number of miles driven by automobile owners in the United States. Many things might affect total miles driven. More or fewer people may be driving. This number, in turn, can be affected by changes in the driving age, by population growth, or by changes in state laws. Other factors might include the price of gasoline, the household's income, the number and age of children in the household, the distance from home to work, the location of shopping facilities, and the availability and quality of public transport. When any of these variables change, the members of the household may

drive more or less. If changes in any of these variables affect large numbers of households across the country, the total number of miles driven will change.

Very often we need to isolate or separate these effects. For example, suppose we want to know the impact on driving of a higher tax on gasoline. This increased tax would raise the price of gasoline at the pump, and this could reduce driving.

To isolate the impact of one single factor, we use the device of **ceteris paribus,** or **all else equal**. We ask, "What is the impact of a change in gasoline price on driving behavior, ceteris paribus, or assuming that nothing else changes?" If gasoline prices rise by 10 percent, how much less driving will there be, assuming no simultaneous change in anything else—that is, assuming that income, number of children, population, laws, and so on, all remain constant? Using the device of ceteris paribus is one part of the process of abstraction. In formulating economic theory, the concept helps us simplify reality to focus on the relationships that interest us.

ceteris paribus, or **all else equal** A device used to analyze the relationship between two variables while the values of other variables are held unchanged.

Expressing Models in Words, Graphs, and Equations

Consider the following statements: Lower airline ticket prices cause people to fly more frequently. Higher gasoline prices cause people to drive less and to buy more fuel-efficient cars. By themselves, these observations are of some interest. But for a firm, government, or an individual to make good decisions, oftentimes they need to know more. How much does driving fall when prices rise? Quantitative analysis is an important part of economics as well. Throughout this book, we will use both graphs and equations to capture the quantitative side of our economic observations and predictions. The appendix to this chapter reviews some graphing techniques.

Cautions and Pitfalls

In formulating theories and models, it is especially important to separate causation from correlation.

What Is Really Causal? In much of economics, we are interested in cause and effect. But cause and effect are often difficult to figure out. Recently, many people in the United States have begun to worry about consumption of soda and obesity. Some areas have begun taxing soda, trying to raise the price so that people will drink less of it. Is this working? Answering this question turns out to be hard. Suppose we see that one city raises the tax and at more or less the same time, soda consumption falls. Did the increased tax and price really *cause* all or most of the change in behavior? Or perhaps the city that voted the soda tax increase is more health conscious than its neighbors and it is that health consciousness that accounts for both the town's decision to raise taxes *and* its reduction in soda purchases. In this case, raising taxes in the neighboring towns will not necessarily reduce soda consumption. Sorting out causality from correlation is not always easy, particularly when one wants a quantitative answer to a question.

In our everyday lives, we often confuse causality. When two events occur in a sequence, it seems natural to think A caused B. I walked under a ladder and subsequently stubbed my toe. Did the ladder cause my bad luck? Most of us would laugh at this. But everyday we hear stock market analysts make a similar causal jump. "Today the Dow Jones industrial average rose 100 points on heavy trading due to progress in talks between Israel and Syria." How do they know this? Investors respond to many news events on any given day. Figuring out which one, if any, causes the stock market to rise is not easy. The error of inferring causality from two events happening one after the other is called the **post hoc, ergo propter hoc** fallacy ("after this, therefore because of this"). The *Economics in Practice* box describes a causality confusion in looking at the effect of part-time jobs on academic performance.

post hoc, ergo propter hoc Literally, "after this (in time), therefore because of this." A common error made in thinking about causation: If Event A happens before Event B, it is not necessarily true that A caused B.

Testing Theories and Models: Empirical Economics

In science, a theory is rejected when it fails to explain what is observed or when another theory better explains what is observed. The collection and use of data to test economic theories is called **empirical economics**.

empirical economics The collection and use of data to test economic theories.

Numerous large data sets are available to facilitate economic research. For example, economists studying the labor market can now test behavioral theories against the actual working experiences of thousands of randomly selected people who have been surveyed continuously since the 1960s. Macroeconomists continuously monitoring and studying the behavior of the national economy at the National Bureau of Economic Research (NBER) analyze thousands of items of data, collected by both government agencies and private companies, over the Internet. Firms like Google, Uber, and Amazon have an enormous amount of data about individual consumers that they analyze with the help of PhD economists to understand consumers' buying behavior and improve the profitability of their businesses. In doing this analysis, economists have learned to be especially careful about causality issues.

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ECONOMICS IN PRACTICE

Does Your Part-Time Job Affect Your Academic Performance?

In order to cover the cost of living, getting a part-time job is a common practice among students; however, parents are concerned that combining paid work with academics may jeopardize their child's performance. Evidence from across institutions indicates that a student working a part-time job is more likely to feel stressed, lack a social life, and face difficulties in completing assignments if the job is strenuous or poorly managed. This poses an interesting question on causality: does part-time employment cause poor academic performance or does a student's academic predisposition cause her to choose such a job?

Given that getting a part-time job is often born out of necessity, determining causality in situations that are only partially a matter of choice is difficult. However, some studies, which include the effects of part-time jobs on academic performance, do solve the causality conundrum.



In Canada, university students can choose from various part-time jobs and most students do not complete non-course related work by their own choice. To test the effects of various types of part-time jobs, a study was conducted by Marsha Barber, a professor at Ryerson University, Canada, and Julia Levitan, a psychology student at the University of Guelph, Canada. They used data on final-year journalism students' academic achievements and well-being at an urban Canadian university, the reason for getting a part-time job, and the number of working hours.[1] The data indicated strong negative employment effects—predominant in students who worked for more than 20 hours a week—on the academic performance, learning engagement, motivation, and well-being of the student. In contrast, students at other universities are randomly assigned various course related part-time jobs within their schools. To test the employment effects on academic performance in such a situation, Mikhail Kouliavtsev, a professor at Stephen F. Austin State University, Texas, used a distinctive student dataset compiled over five years in a business course taught at a U.S. comprehensive regional public university. Mikhail observed that working smartly does have positive effects on academic performance, while longer working hours have a significant negative effect.[2] The bottom line: Choose your part-time jobs wisely!

CRITICAL THINKING

1. Do you think the academic performance of university juniors changes when they become seniors if they continue to pursue part-time employment? Why or why not?

[1] Marsha Barber and Julia Levitan, "Balancing the Books," *The International Journal of Learning in Higher Education,* 2015, 21, 13–19.
[2] Mikhail Kouliavtsev, "The Impact of Employment and Extracurricular Involvement on Undergraduates' Performance in a Business Statistics Course," *Journal of Economics and Economic Education Research,* September 2013, 53–66.

In the natural sciences, controlled experiments, typically done in the lab, are a standard way of testing theories. In recent years, economics has seen an increase in the use of experiments, both in the field and in the lab, as a tool to test its theories. One economist, John List of Chicago, tested the effect on prices of changing the way auctions for rare baseball cards were run by sports memorabilia dealers in trade shows. (The experiment used a standard Cal Ripken Jr. card.) Another economist, Keith Chen of UCLA, has used experiments with monkeys to investigate the deeper biological roots of human decision making.

Economic Policy MyLab Economics Concept Check

Economic theory helps us understand how the world works, but the formulation of *economic policy* requires a second step. We must have objectives. What do we want to change? Why? What is good and what is bad about the way the system is operating? Can we make it better?

Such questions force us to be specific about the grounds for judging one outcome superior to another. What does it mean to be better? Four criteria are frequently applied in judging economic outcomes:

1. Efficiency
2. Equity

3. Growth
4. Stability

Efficiency In physics, "efficiency" refers to the ratio of useful energy delivered by a system to the energy supplied to it. An efficient automobile engine, for example, is one that uses a small amount of fuel per mile for a given level of power.

In economics, **efficiency** means *allocative efficiency*. An efficient economy is one that produces what people want at the least possible cost. If the system allocates resources to the production of goods and services that nobody wants, it is inefficient. If all members of a particular society were vegetarians and somehow half of all that society's resources were used to produce meat, the result would be inefficient.

efficiency The condition in which the economy is producing what people want at the least possible cost.

The clearest example of an efficient change is a voluntary exchange. If you and I each want something that the other has and we agree to exchange, we are both better off and no one loses. When a company reorganizes its production or adopts a new technology that enables it to produce more of its product with fewer resources, without sacrificing quality, it has made an efficient change. At least potentially, the resources saved could be used to produce more of something else.

Inefficiencies can arise in numerous ways. Sometimes they are caused by government regulations or tax laws that distort otherwise sound economic decisions. Suppose that land in Ohio is best suited for corn production and that land in Kansas is best suited for wheat production. A law that requires Kansas to produce only corn and Ohio to produce only wheat would be inefficient. If firms that cause environmental damage are not held accountable for their actions, the incentive to minimize those damages is lost and the result is inefficient.

Equity While efficiency has a fairly precise definition that can be applied with some degree of rigor, **equity** (fairness) lies in the eye of the beholder. To many, fairness implies a more equal distribution of income and wealth. For others, fairness involves giving people what they earn. In 2013, French economist Thomas Piketty's popular new book *Capital in the Twenty-First Century*, brought new historical data to our attention on the extent of inequality across the Western world. More recent work by Raj Chetty of Stanford University has greatly improved our understanding of economic mobility in the United States, documenting the extent to which parental and adult children's incomes are correlated.

equity Fairness.

Growth As the result of technological change, the building of machinery, and the acquisition of knowledge, societies learn to produce new goods and services and to produce old ones better. In the early days of the U.S. economy, it took nearly half the population to produce the required food supply. Today less than 2 percent of the country's population works in agriculture.

When we devise new and better ways of producing the goods and services we use now and when we develop new goods and services, the total amount of production in the economy increases. **Economic growth** is an increase in the total output of an economy. If output grows faster than the population, output per person rises and standards of living increase. Rural and agrarian societies become modern industrial societies as a result of economic growth and rising per capita output.

economic growth An increase in the total output of an economy. Growth occurs when a society acquires new resources or when it learns to produce more using existing resources.

Some policies discourage economic growth, and others encourage it. Tax laws, for example, can be designed to encourage the development and application of new production techniques. Research and development in some societies are subsidized by the government. Building roads, highways, bridges, and transport systems in developing countries may speed up the process of economic growth. If businesses and wealthy people invest their wealth outside their country rather than in their country's industries, growth in their home country may be slowed.

Stability Economic **stability** refers to the condition in which national output is growing steadily, with low inflation and full employment of resources. During the 1950s and 1960s, the U.S. economy experienced a long period of relatively steady growth, stable prices, and low unemployment. The decades of the 1970s and 1980s, however, were not as stable. The United States experienced two periods of rapid price inflation (more than 10 percent) and two periods of severe unemployment. In 1982, for example, 12 million people (10.8 percent of the workforce) were looking for work. The beginning of the 1990s was another period of instability, with a recession occurring in 1990–1991. In 2008–2009, much of the world, including the United States, experienced a large contraction in output and rise in unemployment. The period since 2009 in the United States has been one of modest growth and falling unemployment. The causes of instability and the ways in which governments have attempted to stabilize the economy are the subject matter of macroeconomics.

stability A condition in which national output is growing steadily, with low inflation and full employment of resources.

Troy University Resources

1.4 LEARNING OBJECTIVE

Begin to get a sense of the many ways economics touches one's life.

An Invitation

This chapter has prepared you for your study of economics. The first part of the chapter invited you into an exciting discipline that deals with important issues and questions. You cannot begin to understand how a society functions without knowing something about its economic history and its economic system.

The second part of the chapter introduced the method of reasoning that economics requires and some of the tools that economics uses. We believe that learning to think in this powerful way will help you better understand the world.

As you proceed, it is important that you keep track of what you have learned in previous chapters. This book has a plan; it proceeds step-by-step, each section building on the last. It would be a good idea to read each chapter's table of contents at the start of each chapter and scan each chapter before you read it to make sure you understand where it fits in the big picture.

1.5 LEARNING OBJECTIVE

Describe economics as a career and the key skills you can learn from studying economics.

Economic Skills and Economics as a Career

In this book, we will explore economic principles that you will find very useful in understanding what is happening in the world of economics and business and in your everyday life. Individuals use economic principles to improve how they make important decisions, such as what career to pursue or what financial investment to make. Managers in businesses use economic principles to improve how they make important decisions, such as what prices to charge for their products or whether to invest in new software. Government policymakers use economic principles to make decisions, such as how to allocate additional funds to research in certain areas. Whether or not you pursue a career in economics, you can still benefit from the skills learned by taking economics classes.

SUMMARY

1. *Economics* is the study of how individuals and societies choose to use the scarce resources that nature and previous generations have provided.

1.1 WHY STUDY ECONOMICS? *p. 28*

2. There are many reasons to study economics, including (a) to learn a way of thinking, (b) to understand society, and (c) to be an informed citizen.
3. The best alternative that we forgo when we make a choice or a decision is the *opportunity cost* of that decision.

1.2 THE SCOPE OF ECONOMICS *p. 31*

4. *Microeconomics* deals with the functioning of individual markets and industries and with the behavior of individual decision-making units: business firms and households.
5. *Macroeconomics* looks at the economy as a whole. It deals with the economic behavior of aggregates—national output, national income, the overall price level, and the general rate of inflation.
6. Economics is a broad and diverse discipline with many special fields of inquiry. These include economic history, international economics, and urban economics.

1.3 THE METHOD OF ECONOMICS *p. 34*

7. Economics asks and attempts to answer two kinds of questions: positive and normative. *Positive economics* attempts to understand behavior and the operation of economies without making judgments about whether the outcomes are good or bad. *Normative economics* looks at the results of economic behavior and asks whether they are good or bad and whether they can be improved.
8. An economic *model* is a formal statement of an economic theory. Models simplify an abstract from reality.
9. It is often useful to isolate the effects of one variable on another while holding "all else constant." This is the device of *ceteris paribus.*
10. Models and theories can be expressed in many ways. The most common ways are in words, in graphs, and in equations.
11. Figuring out causality is often difficult in economics. Because one event happens before another, the second event does not necessarily happen as a result of the first. To assume that "after" implies "because" is to commit the fallacy of *post hoc, ergo propter hoc.*
12. *Empirical economics* involves the collection and use of data to test economic theories. In principle, the best model is the one that yields the most accurate predictions.
13. To make policy, one must be careful to specify criteria for making judgments. Four specific criteria are used most often in economics: *efficiency, equity, growth,* and *stability.*

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